Atmel AT89S52

The **AT89S52** is a classic, high-performance 8-bit microcontroller from the 8051 family, manufactured by Atmel (now part of Microchip). It remains a popular choice for education and simple embedded systems due to its simplicity and the inclusion of **In-System Programming (ISP)**.

Key Specifications

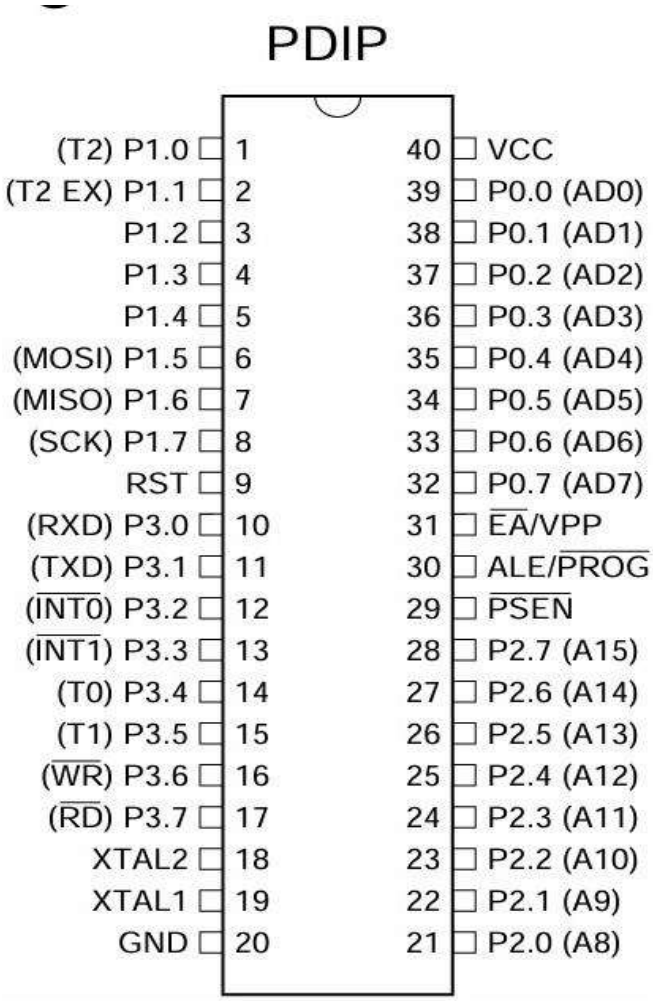
The AT89S52 is essentially an upgraded version of the standard 8051, offering more memory and an extra timer.

- **Core:** 8-bit CISC (Complex Instruction Set Computer).
- **Flash Memory:** 8KB of ISP (In-System Programmable) Flash.
- **RAM:** 256 bytes of internal RAM.
- **Timers:** Three 16-bit timers/counters.
- **I/O Pins:** 32 programmable I/O lines (organized as four 8-bit ports).
- **Interrupts:** 8 interrupt sources.
- **Clock Frequency:** 0Hz to 33MHz.
- **Operating Voltage:** 4.0V to 5.5V.

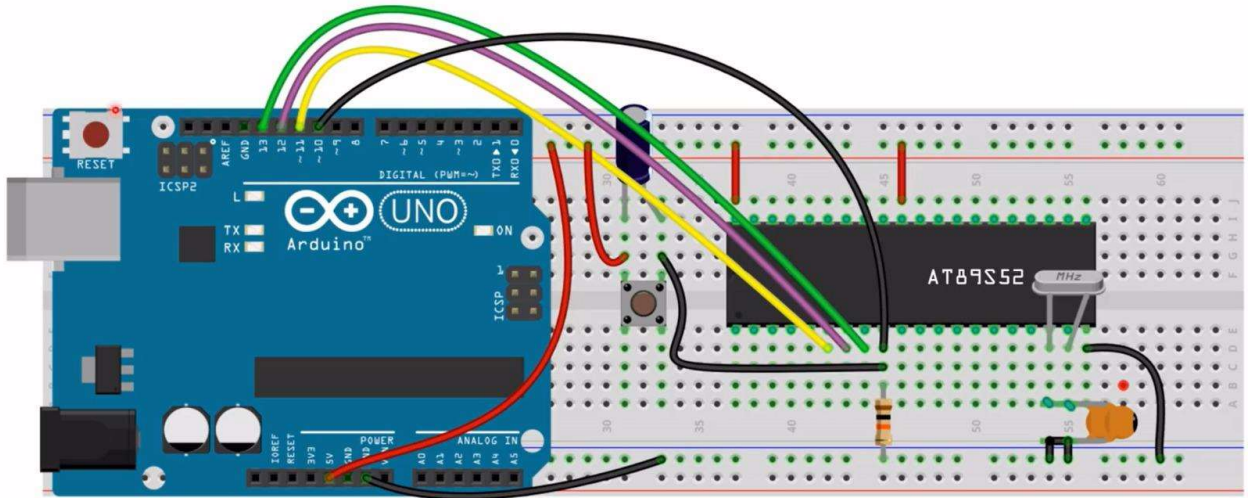
#1.Pin Configuration

The microcontroller comes in a **40-pin DIP** (Dual In-line Package), making it very easy to use on breadboards.

- **Port 0 (Pins 32-39):** 8-bit bidirectional I/O port. It also serves as the low-order address and data bus during access to external memory.
- **Port 1 (Pins 1-8):** 8-bit bidirectional I/O port. Pins P1.5, P1.6, and P1.7 are used for ISP programming (MOSI, MISO, and SCK).
- **Port 2 (Pins 21-28):** 8-bit bidirectional I/O port. It also provides the high-order address byte during external memory fetches.
- **Port 3 (Pins 10-17):** Dual-purpose port. Besides I/O, these pins handle functions like UART (serial comms), external interruption, and timer inputs.
- **RST (Pin 9):** Reset input. A high level on this pin for two machine cycles while the oscillator is running will reset the device.



#2. Connection (Arduino UNO ↔ AT89S52)



#2.1 Connection Table: -

🧠 Data Communication (Programming / IO lines)

Arduino UNO Pin	AT89S52 Pin	Purpose
D13	P1.7	Data / Control
D12	P1.6	Data / Control
D11	P1.5	Data / Control
D10	RESET	

(These are general-purpose connections; exact function depends on your code — SPI/parallel control)

⚡ Power Supply

Arduino UNO	AT89S52	Purpose
5V	Pin 40 (VCC)	Power supply
	Pin 31(VPP)	Power Supply
GND	Pin 20 (GND)	Ground

🔁 Reset Circuit

Component	Connection
Push Button	One side → VCC
	Other side → Reset Pin (Pin 9 of AT89S52)
Resistor (~10kΩ)	Between Reset Pin and GND

⏱ Crystal Oscillator

Component	AT89S52 Pin
Crystal (11.0592 MHz typical)	Pin 18 (XTAL2)
	Pin 19 (XTAL1)
Capacitor 1 (~33pF)	Pin 18 → GND
Capacitor 2 (~33pF)	Pin 19 → GND

🔦 Additional Components

Component	Connection
Capacitor (orange)	Between VCC & GND (filtering)
LED (if present)	Through resistor to any port pin

#3. Procedure: Interfacing Arduino UNO with AT89S52 Circuit

◇ Step 1: Setup Power Supply

1. Connect **5V pin of Arduino UNO → VCC (Pin 40) of AT89S52**
2. Connect **GND of Arduino → GND (Pin 20) of AT89S52**
3. Ensure **common ground** across the breadboard

◇ Step 2: Place AT89S52 on Breadboard

4. Insert the AT89S52 IC in the center of the breadboard
5. Make sure both sides of pins are accessible

◇ Step 3: Connect Crystal Oscillator

6. Connect crystal between:
    a. **Pin 18 (XTAL2)**
    b. **Pin 19 (XTAL1)**
7. Connect two capacitors (~33pF each):
    a. One from Pin 18 → GND
    b. One from Pin 19 → GND

👉 This provides the clock signal to the microcontroller.

◇ Step 4: Build Reset Circuit

8. Connect **Push Button**:
    a. One side → VCC
    b. Other side → **Pin 9 (RESET)**
9. Connect **10kΩ resistor**:

a. Between RESET pin → GND

👉 This ensures proper startup/reset of AT89S52.

◇ Step 5: Connect Arduino to AT89S52 (Data Lines)

10. Connect Arduino digital pins:

- **D13 → P1.0 (Pin 1)**
- **D12 → P1.1 (Pin 2)**
- **D11 → P1.2 (Pin 3)**
- **D10 → P1.3 (Pin 4)**

👉 These act as control/data signals (depends on your program).

◇ Step 6: Add Decoupling Capacitor

11. Place a capacitor (0.1µF–10µF):

- Between VCC and GND

👉 Helps in noise filtering and stable operation.

◇ Step 7: Verify Connections

12. Check:

- No loose wires
- No short circuits
- Correct polarity of components

◇ Step 8: Power ON the Circuit

13. Connect Arduino to laptop via USB
14. Arduino will supply power to AT89S52

◇ Step 9: Upload Arduino Code

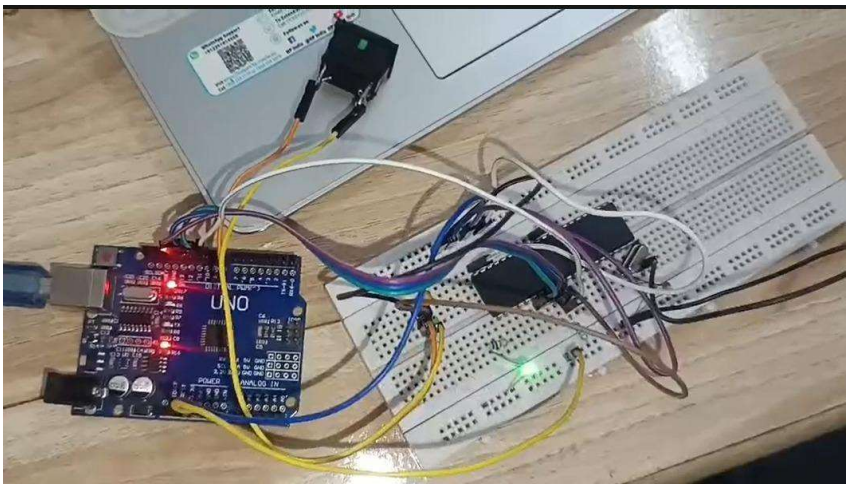
15. Write and upload code to Arduino for:

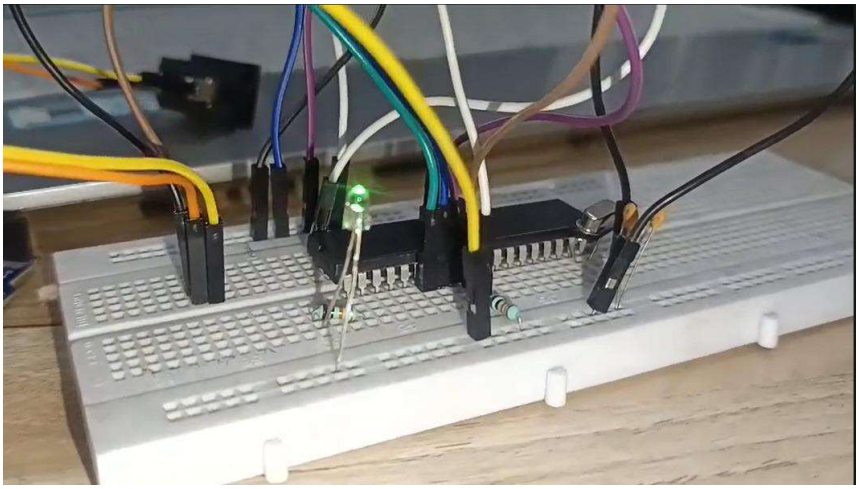
- Sending signals
- Controlling AT89S52 pins

◇ Step 10: Test the Circuit

16. Press reset button → AT89S52 should restart
17. Observe output (LED / signal behavior based on code)

#4. OUTPUT OF THESE CONNECTION RESULTS IN: -





#5. Result: -

- The **AT89S52 microcontroller circuit** was successfully assembled on the breadboard with proper power supply, reset circuit, and crystal oscillator.
- The **Arduino UNO** was able to provide input/control signals to the AT89S52 through connected digital pins.
- On powering the circuit, the AT89S52 responded correctly to the reset button and operated as expected based on the given input signals.
- The clock circuit (crystal + capacitors) ensured stable operation of the microcontroller.
- No abnormal behavior such as overheating, noise, or unstable output was observed.

👉 Hence, the interfacing between Arduino and AT89S52 was verified successfully.

Report by -

Radhika Gupta

Param Dohele

Ravi Singh